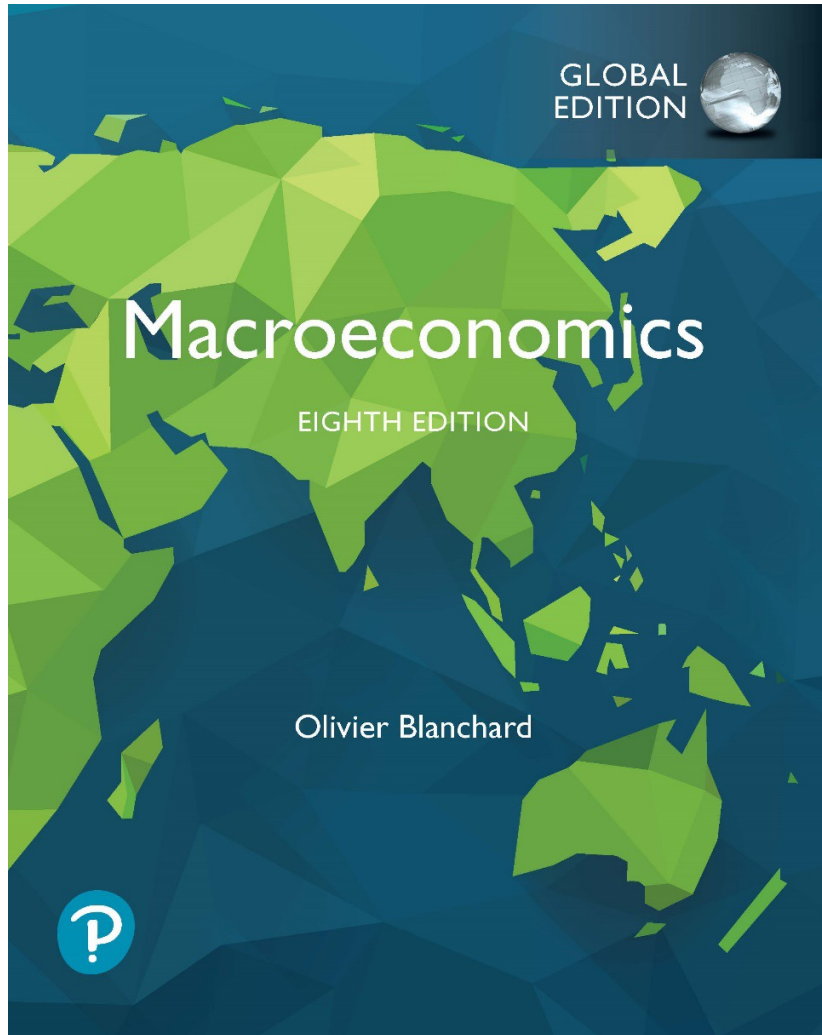


Macroeconomics

Eighth Edition, Global Edition



Chapter 2

A Tour of the Book

Chapter 2 Outline

A Tour of the Book

2.1 Aggregate Output

2.2 The Unemployment Rate

2.3 The Inflation Rate

2.4 Output, Unemployment, and the Inflation Rate: Okun's Law and the Phillips Curve

2.5 The Short Run, the Medium Run, and the Long Run

2.6 A Tour of the Book

APPENDIX The Construction of Real GDP and Chain-Type Indexes

A Tour of the Book

- The words *output*, *unemployment*, and *inflation* appear daily in newspapers and on the evening news.
- In this chapter, we define these words more precisely.
- The chapter also introduces concepts around which the book is organized: the *short run*, the *medium run*, and the *long run*.

2.1 Aggregate Output (1 of 11)

- **National income and product accounts** were developed at the end of World War II as measures of aggregate output.
- The measure of **aggregate output** is called gross domestic product (**GDP**).
- How would you define aggregate output in the economy?

2.1 Aggregate Output (2 of 11)

Steel Company (Firm 1)		Car Company (Firm 2)	
Revenues from sales	\$100	Revenues from sales	\$200
Expenses	\$80	Expenses	\$170
Wages	\$80	Wages	\$70
		Steel purchases	\$100
Profit	\$20	Profit	\$30

- Consider an economy with two firms, Firm 1 and Firm 2.
- Is aggregate output the sum of the values of all goods produced, i.e., \$300? Or just the value of cars, i.e., \$200?
- Steel is an **intermediate good**, which is a good used in the production of another good.

$$\text{Income of factors} = \text{wage} + \text{profit} = (70 + 80) + (20 + 30) = 200 \text{ \$}$$

$$\text{Valued-added: } 100 \text{ from steel company} + 100 \text{ from car company} = 200$$

2.1 Aggregate Output (3 of 11)

1. GDP is the ^{market value} value of the **final goods and services** produced in the economy during a given period.
 - We want to count only **final goods**, not intermediate goods.
 - If we merge the two firms in the previous example, the revenues of the new firm equal \$200.

value of intermediate goods include in value of final goods

2.1 Aggregate Output (4 of 11)

2 GDP is the sum of value added in the economy during a given period.

- The **value added** by a firm is the value of its production minus the value of the intermediate goods used in production.
 - In the two-firm example, the value added equals \$100 + \$100 = \$200.
- So far, we have looked at GDP from the *production side*.

Steel and Car Company	
Revenues from sales	\$200
Expenses (wages)	\$150
Profit	\$50

2.1 Aggregate Output (5 of 11)

3. GDP is the **sum of incomes in the economy** during a given period.

- Aggregate production and aggregate income are always equal.
 - From the *income side*, valued added in the two-firm example is equal to the sum of *labor income* (\$150) and *capital or profit* income (\$50), i.e., \$200.
- From the *production side*: GDP equals the value of the final goods and services produced in the economy during a given period.
 - Also from the *production side*: GDP is the sum of value added in the economy during a given period.
 - From the *income side*: GDP is the sum of incomes in the economy during a given period.

2.1 Aggregate Output (6 of 11)

- **Nominal GDP** is the sum of the quantities of final goods produced times their **current price**.
- Nominal GDP increases for two reasons:
 - The production of most goods increases over time
 - The price of most goods increases over time
- Our goal is to measure production and its change over time.
- **Real GDP** is the sum of quantities of final goods times **constant (not current) prices**. 經濟成長要看的指標
價格不變，數量變動

2.1 Aggregate Output (7 of 11)

- **Example:**

Year	Quantity of Cars	Price of Cars	Nominal GDP	Real GDP (in 2012 dollars)
2011	10	\$20,000	\$200,000	\$240,000
2012	12	\$24,000	\$288,000	\$288,000
2013	13	\$26,000	\$338,000	\$312,000

Base year

constant
price

- Real GDP in 2011 (in 2012 dollars) = 10 cars x \$24,000 per car = \$240,000.
- Real GDP in 2012 (in 2012 dollars) = 12 cars x \$24,000 per car = \$288,000.
- Real GDP in 2013 (in 2012 dollars) = 13 cars x \$24,000 per car = \$312,000.

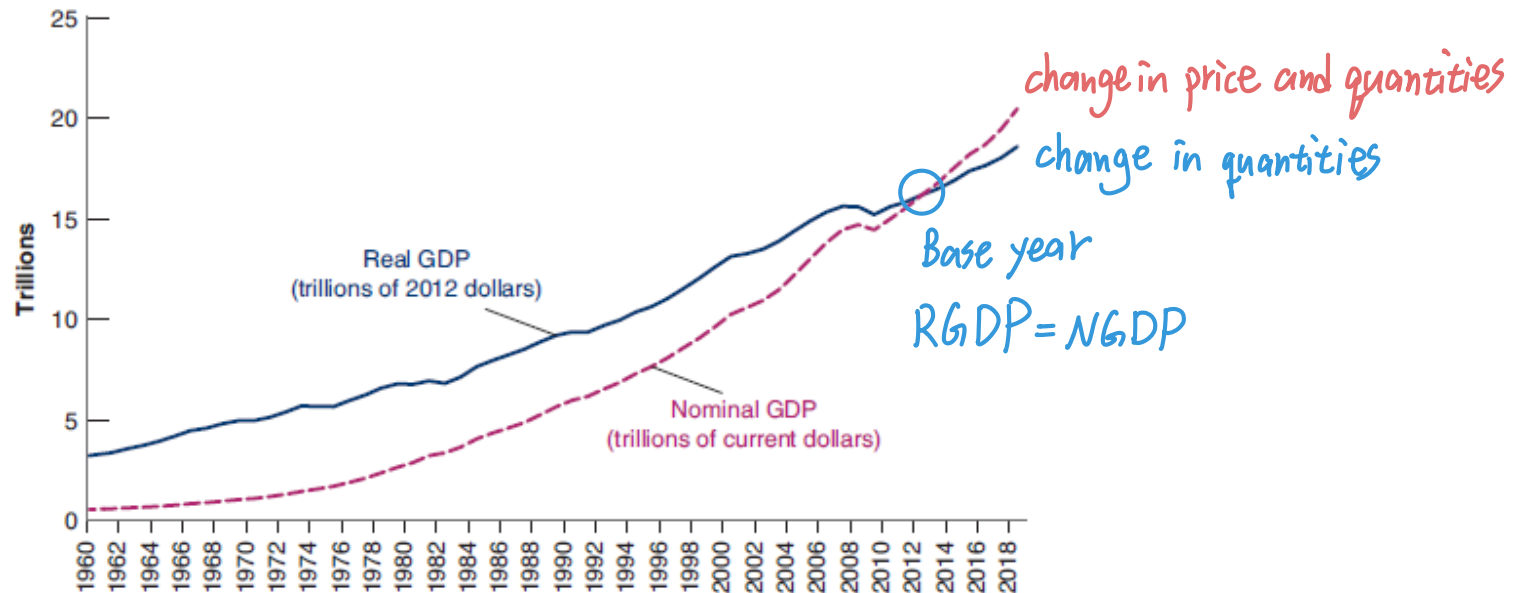
2.1 Aggregate Output (8 of 11)

- For more than one good, relative prices of the goods are natural weights for constructing the weighted average of the output of all final goods.
- **Real GDP in chained (2012) dollars** reflects relative prices that change over time.
- The year used to construct prices is called the *base year*.

2.1 Aggregate Output (9 of 11)

Figure 2.1 Nominal and Real U.S. GDP, 1960–2018

From 1960 and 2018, nominal GDP increased by a factor of 38. Real GDP increased by a factor of about 5.7.



Source: FRED. Series GDPC, GDP.

2.1 Aggregate Output (10 of 11)

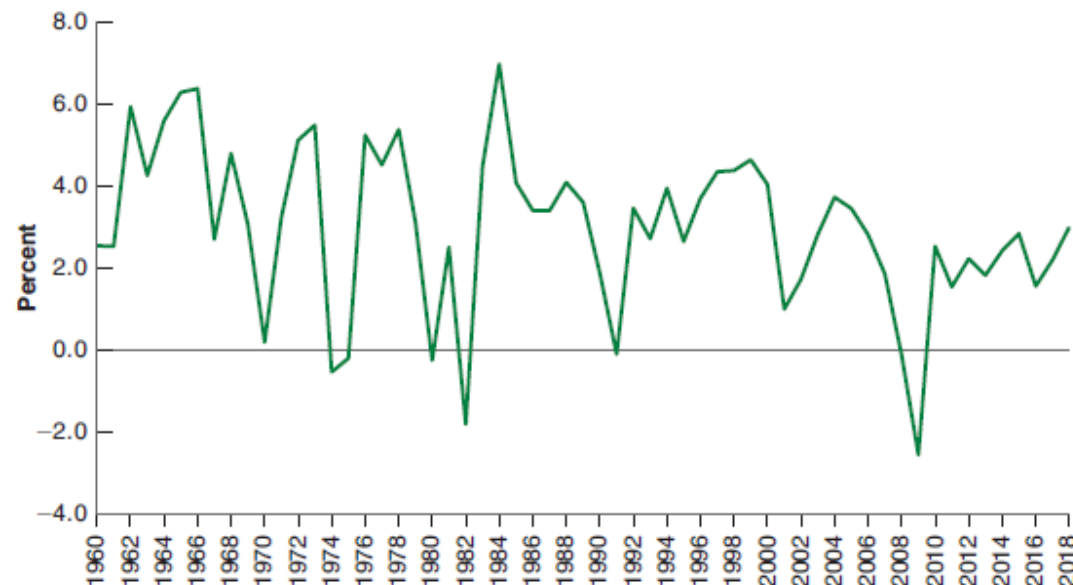
- $\$Y$ Nominal GDP is also called **dollar GDP**, or **GDP in current dollars**.
- Y Real GDP is also called **GDP in terms of goods**, **GDP in constant dollars**, **GDP adjusted for inflation**, or **GDP in chained (2012) dollars**, or **GDP in 2012 dollars**.
 - GDP will refer to *real GDP*.
 - Y_t will denote *real GDP in year t* .
 - Nominal GDP and variables in current dollars will be denoted by a dollar sign in front of them, e.g., $\$Y_t$

1.2 The United States (4 of 6)

Figure 2.2 Growth Rate of U.S. GDP, 1960–2018

GDP growth in year t is $(Y_t - Y_{t-1})/Y_{t-1}$.

Since 1960, the U.S. economy has gone through a series of expansions, interrupted by short recessions. The 2008–2009 recession was the most severe recession in the period from 1960 to 2018.



Source: Calculated using series GDPC in Figure 2.1.

FOCUS: Real GDP, Technological Progress, and the Price of Computers

- The Department of Commerce deals with changes in the quality of existing goods like computers with an approach called **hedonic pricing**, which treats goods as providing a collection of characteristics.
- The quality of new laptops (computing services) has increased on average by 20% a year since 1999.
- The dollar price of a typical laptop has also declined by about 7% a year since 1999.
- This implies that laptops' quality-adjusted price has fallen at an average rate of $20\% + 7\% = 27\%$ per year.

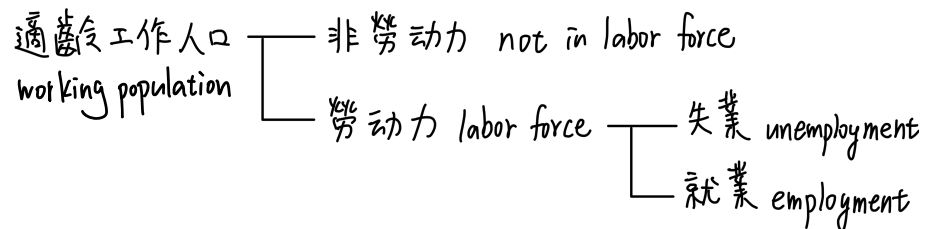
2.2 The Unemployment Rate (1 of 6)

- **Employment** is the number of people who have a job.
- **Unemployment** is the number of people who do not have a job but are looking for one.
- The **labor force** is the sum of employment and unemployment.

$$Y = F(K, L)$$

$$L = N + U$$

labor force = employment + unemployment



2.2 The Unemployment Rate (2 of 6)

- The **unemployment rate** is the ratio of the number of people who are unemployed to the number of people in the labor force.

efficiency lost

$$u = \frac{U}{L}$$

unemployment rate = unemployment / labor force

2.2 The Unemployment Rate (3 of 6)

- Most rich countries rely on large surveys of households to compute the unemployment rate.
- The U.S. **Current Population Survey (CPS)** relies on interviews of 60,000 households every month.
- A person is unemployed if he or she does not have a job *and has been looking for a job in the last four weeks.*
- Those who do not have a job and are not looking for one are counted as **not in the labor force**.

→ 过去四星期有在找工作
但找不到

2.2 The Unemployment Rate (4 of 6)

- **Discouraged workers** *not in labor force* are those persons who give up looking for a job and so no longer count as unemployed.
- The **participation rate** is the ratio of the labor force to the total population of working age.
- Because of discouraged workers, a higher unemployment rate is typically associated with a lower participation rate.

$$\text{participation rate (LPR)} = \frac{\text{labor force}}{\text{working population (LF + NLF)}}$$

$$\text{unemployment rate} = \frac{U}{LF (U+E)} = \frac{1}{1+\frac{E}{U}}$$

$$U \uparrow \left(\frac{E}{U} \downarrow \right) \Leftrightarrow \text{LPR} \downarrow \left(\frac{NLF}{LF} \uparrow \right)$$

b/c discouraged worker

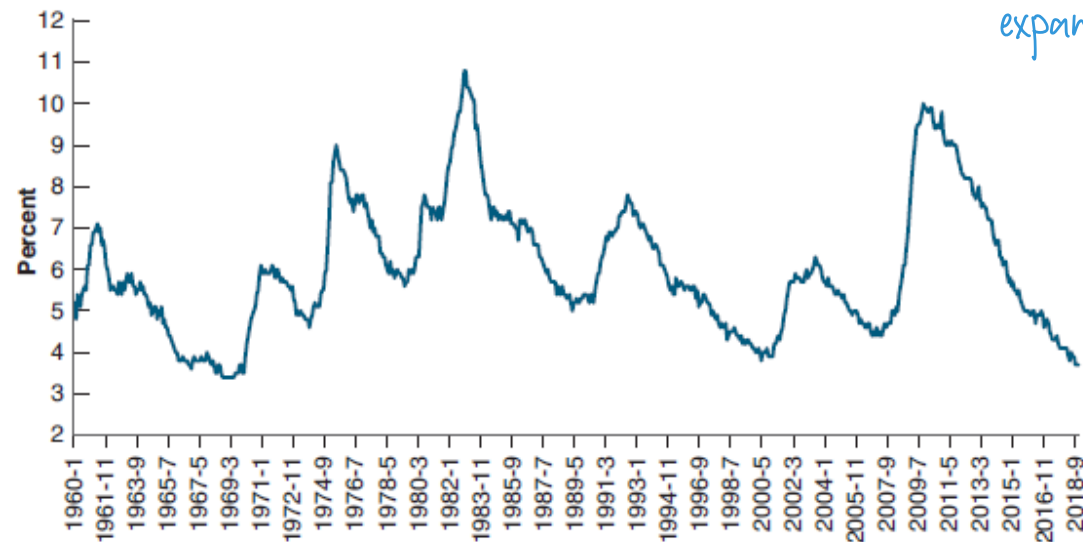
2.2 The Unemployment Rate (5 of 6)

- Why Do Economists Care about Unemployment?
 1. Because of its direct effect on the welfare of the unemployed, especially those remaining unemployed for long periods of time.
 2. It is a signal that the economy is not using its human resources efficiently.
- Very low unemployment can also be a problem as the economy runs into labor shortages.

2.2 The Unemployment Rate (6 of 6)

Figure 2.3 U.S. Unemployment Rate, 1960–2018

- Since 1960, the U.S. unemployment rate has fluctuated between 3 and 11%, going down during expansions and going up during recessions.
- The effect of the recent crisis is highly visible, with the unemployment rate reaching close to 10% in 2010, the highest such rate since the early 1980s.



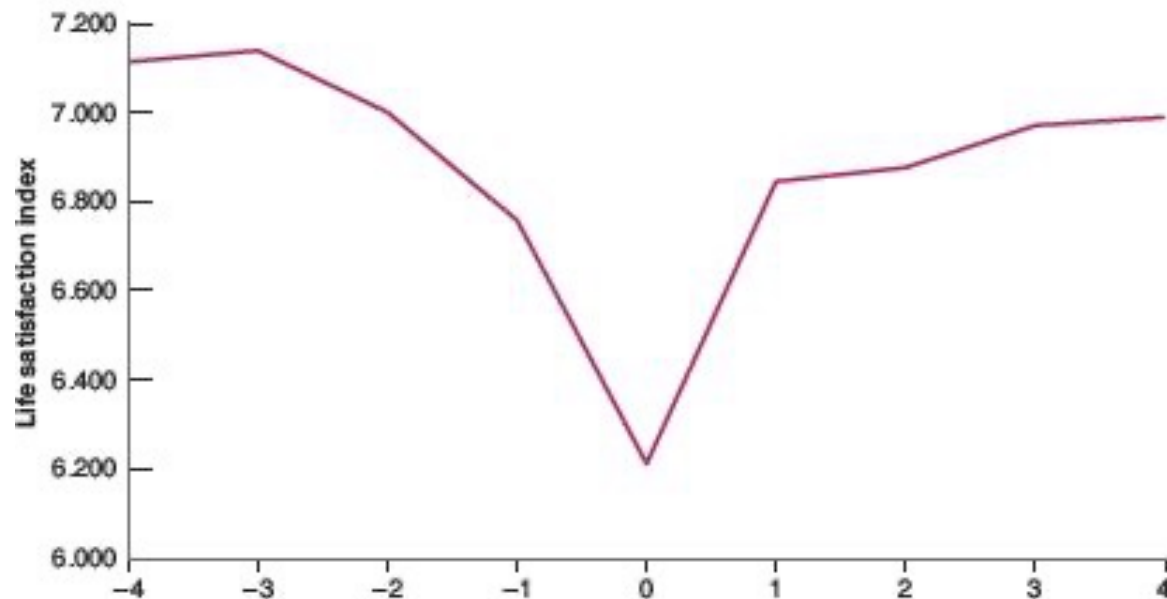
recession → higher u

expansion → lower u

FOCUS: Unemployment and Happiness

- Results of the German Socio-Economic Panel survey suggest that (1) becoming unemployed leads to a large decrease in happiness, (2) happiness declines before the actual unemployment spell, and (3) happiness does not fully recover even four years later.

Figure 1 Effects of Unemployment on Happiness



Source: Winkelmann 2014.

2.3 The Inflation Rate (1 of 8)

- **Inflation** is a sustained rise in the general level of prices—the **price level**.
- The **inflation rate** is the rate at which the price level increases.
- **Deflation** is a sustained decline in the price level (negative inflation rate).

$$(p) \text{ price level} = \frac{NGDP}{RGDP}$$

GDP平減指數

$$\frac{\Delta P}{P} = \frac{P_{t+1} - P_t}{P_t} = \pi_t$$

2.3 The Inflation Rate (2 of 8)

GDP平減指數

- The **GDP deflator** in year t (P_t) is the ratio of nominal GDP to real GDP in year t :

$$P_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} = \frac{\$Y_t}{Y_t}$$

- It is called an **index number** (1 in 2012), which has no economic interpretation.
- The rate of change has a clear interpretation: the rate of inflation.

$$\pi_t = (P_t - P_{t-1})/P_{t-1}$$

inflation rate

2.3 The Inflation Rate (3 of 8)

- Defining the price level as the GDP deflator implies a simple relation between nominal GDP, real GDP, and the GDP deflator:

$$\$Y_t = P_t Y_t$$

- Nominal GDP is equal to the GDP deflator times real GDP.*
- The rate of growth of nominal GDP is equal to the rate of inflation plus the rate of growth of real GDP.

$$\text{GDP growth rate} = \frac{\Delta Y_t}{Y_t} = \frac{Y_{t+1} - Y_t}{Y_t}$$

2.3 The Inflation Rate (4 of 8)

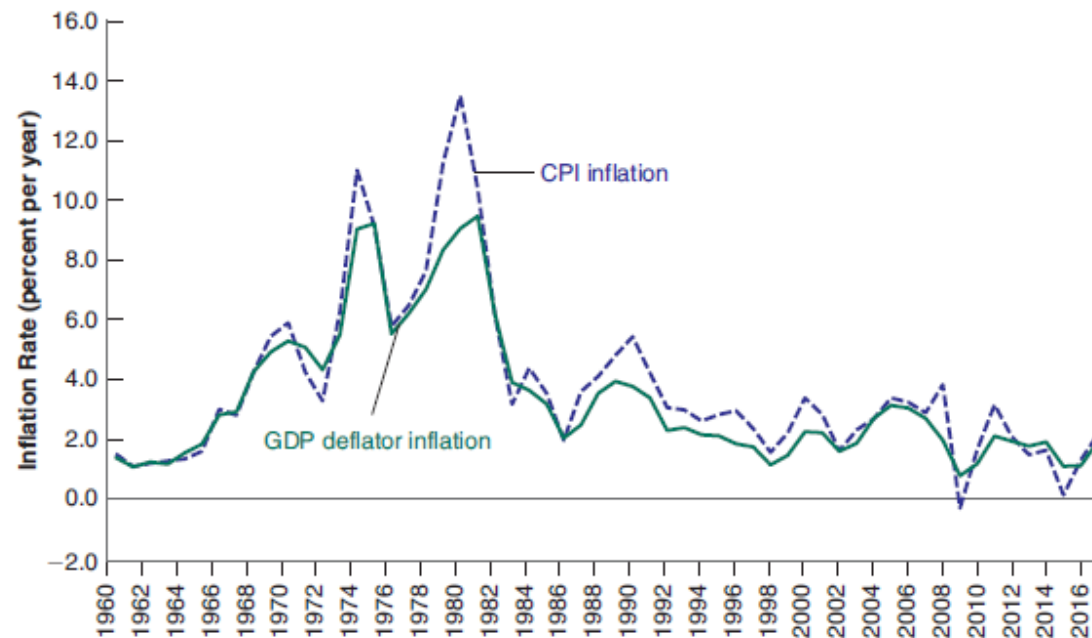
- The set of goods produced in the economy is not the same as the set of goods purchased by consumers because:
 - Some of the goods in GDP are sold not to consumers but to firms, to the government, or to foreigners.
 - Some of the goods bought by consumers are not produced domestically but are imported from abroad.
- The **Consumer Price Index (CPI)** is a measure of the **cost of living**. *daily usage* 消費者物價指數
- The CPI is published monthly by the Bureau of Labor Statistics (BLS), which collects price data for 211 items in 38 cities.
- The CPI gives the cost in dollars of a specific list of goods and services over time.

$$CPI = \frac{\sum P_{ti} q_{oi}}{\sum P_{oi} q_{oi}}$$

2.3 The Inflation Rate (5 of 8)

Figure 2.4 Inflation Rate, Using the CPI and the GDP Deflator, 1960–2018

The inflation rates, computed using either the CPI or the GDP deflator, are largely similar.



Source: FRED: CPIAUCSL and GDPDEF

2.3 The Inflation Rate (6 of 8)

- The CPI and GDP deflator moved together most of the time.
- Exception: In 1979 and 1980, the increase in the CPI was significantly larger than the increase in the GDP deflator due to the price of imported goods increasing relative to the price of domestically produced goods.

2.3 The Inflation Rate (7 of 8)

- *Pure inflation* is proportional increase in all prices and wages.
 - This type of inflation causes only a minor inconvenience as relative prices are unaffected.
 - Real wage (wage measured by goods rather than dollars) would be unaffected.
 - There is no such thing as pure inflation.

$$\frac{P_x}{P_y} \text{ 不变}$$

2.3 The Inflation Rate (8 of 8)

- Why Do Economists Care about Inflation?
 - Inflation affects income distribution when not all prices and wages rise proportionally.
 - Inflation leads to distortions due to uncertainty, some prices that are fixed by law or by regulation, and its interaction with taxation (*bracket creep* in taxes).
- Most economists believe the “best” rate of inflation to be a low and stable rate of inflation between 1 and 4%.

Nominal wage $\uparrow$ $\rightarrow$ higher income tax

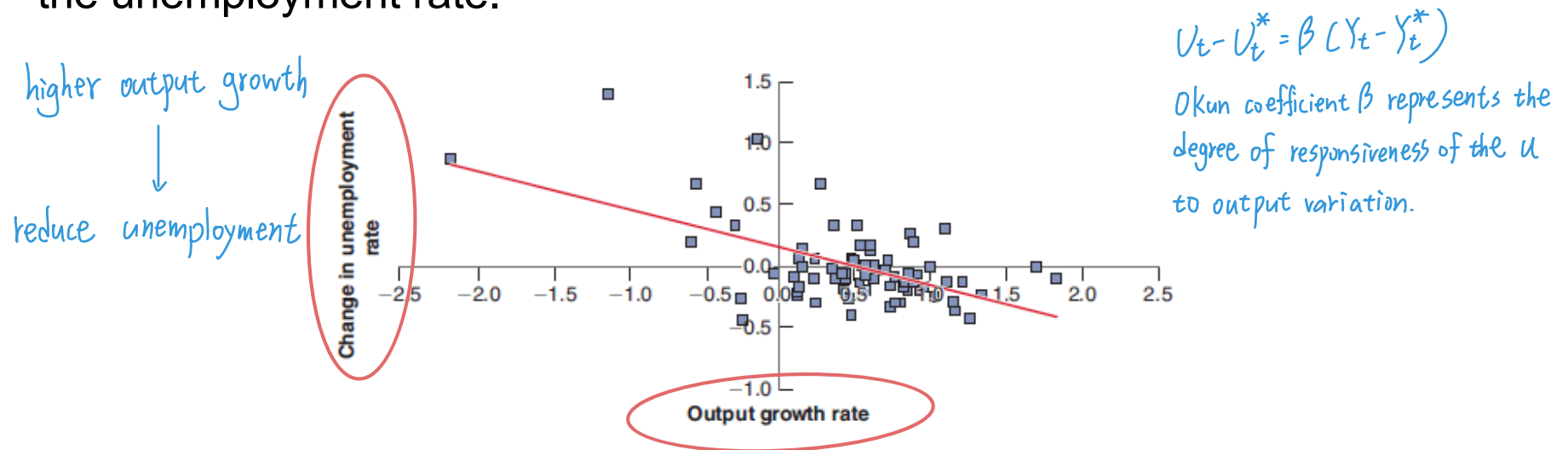
but real wage doesn't change $\rightarrow$ bracket creep

2.4 Output, Unemployment, and the Inflation Rate: Okun's Law and the Phillips Curve (1 of 4)

Figure 2.5 Changes in the Unemployment Rate versus Growth in the United States, 2000 Q1 to 2018 Q4

Output growth that is higher than usual is associated with a reduction in the unemployment rate.

Output growth that is lower than usual is associated with an increase in the unemployment rate.



Source: FRED: Series GDPC, UNRATE.

2.4 Output, Unemployment, and the Inflation Rate: Okun's Law and the Phillips Curve (2 of 4)

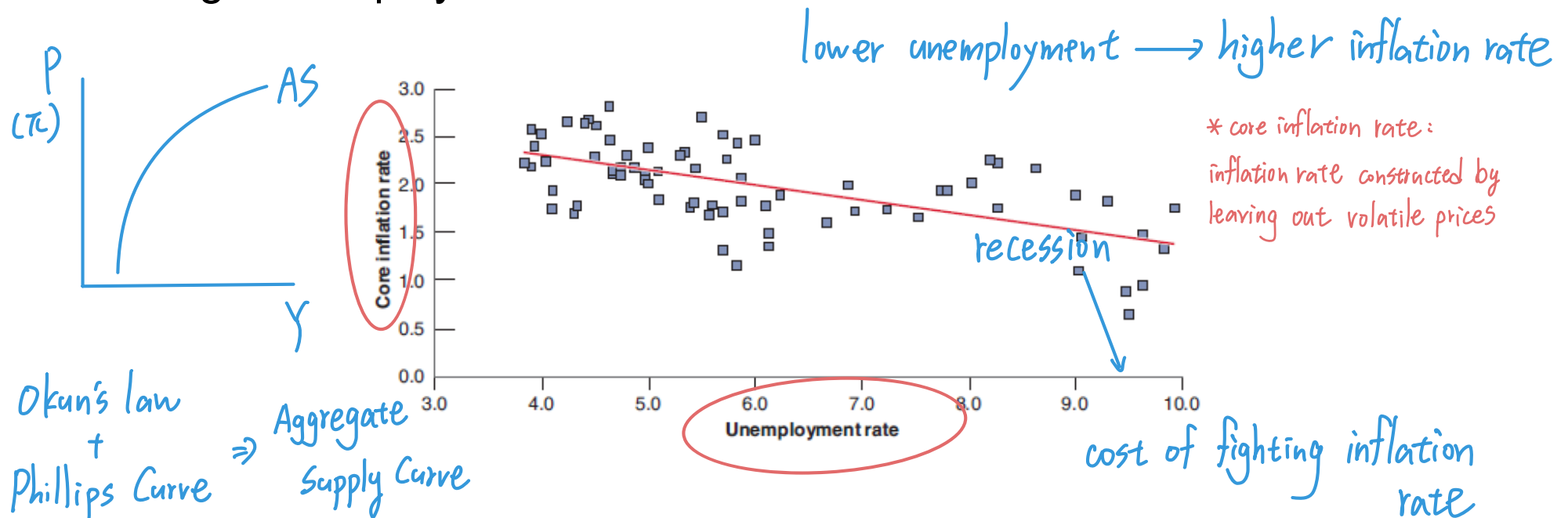
- **Okun's law** is a relation first examined by U.S. economist Arthur Okun.
- In Figure 2-5, the line that best fits the points is downward sloping.
- The slope of the line is -0.3 , which implies that, on average, an increase in the growth rate of 1% decreases the unemployment rate by -0.3% .
 β 值決定需要多大的 output growth
- ✱ The line crosses the horizontal axis where output growth is 0.5% , meaning that it takes a growth rate of 2% to keep unemployment constant.

2.4 Output, Unemployment, and the Inflation Rate: Okun's Law and the Phillips Curve (3 of 4)

Figure 2.6 Changes in the Inflation Rate versus the Unemployment Rate in the United States, 2000 Q4 to 2018 Q4

A low unemployment rate leads to an increase in the inflation rate.

A high unemployment rate leads to a decrease in the inflation rate.



Source: FRED: Series GDPC, CPILFESL.

2.4 Output, Unemployment, and the Inflation Rate: Okun's Law and the Phillips Curve (4 of 4)

- The Phillips curve is a relation first explored in 1958 by New Zealand economist A.W. Phillips.
- Figure 2-6 plots the change in the inflation rate against the unemployment rate, along with the line that best fits the points.
- The line is downward sloping, meaning that higher unemployment leads, on average, to a decrease in inflation, and vice versa.
- When unemployment has been above 5%, inflation has typically been above 2%.

2.5 The Short Run, the Medium Run, and the Long Run

- In the **short run** (e.g., a few years), year-to-year movements in output are primarily driven by movements in demand.
- In the **medium run** (e.g., a decade), the economy tends to return to the level of output determined by supply factors, such as the capital stock, the level of technology, and the size of the labor force.
- In the **long run** (e.g., a few decades or more), the economy depends on its ability to innovate and introduce new technologies, and how much people save, the quality of the country's education system, the quality of the government, and so on.

2.6 A Tour of the Book (1 of 3)

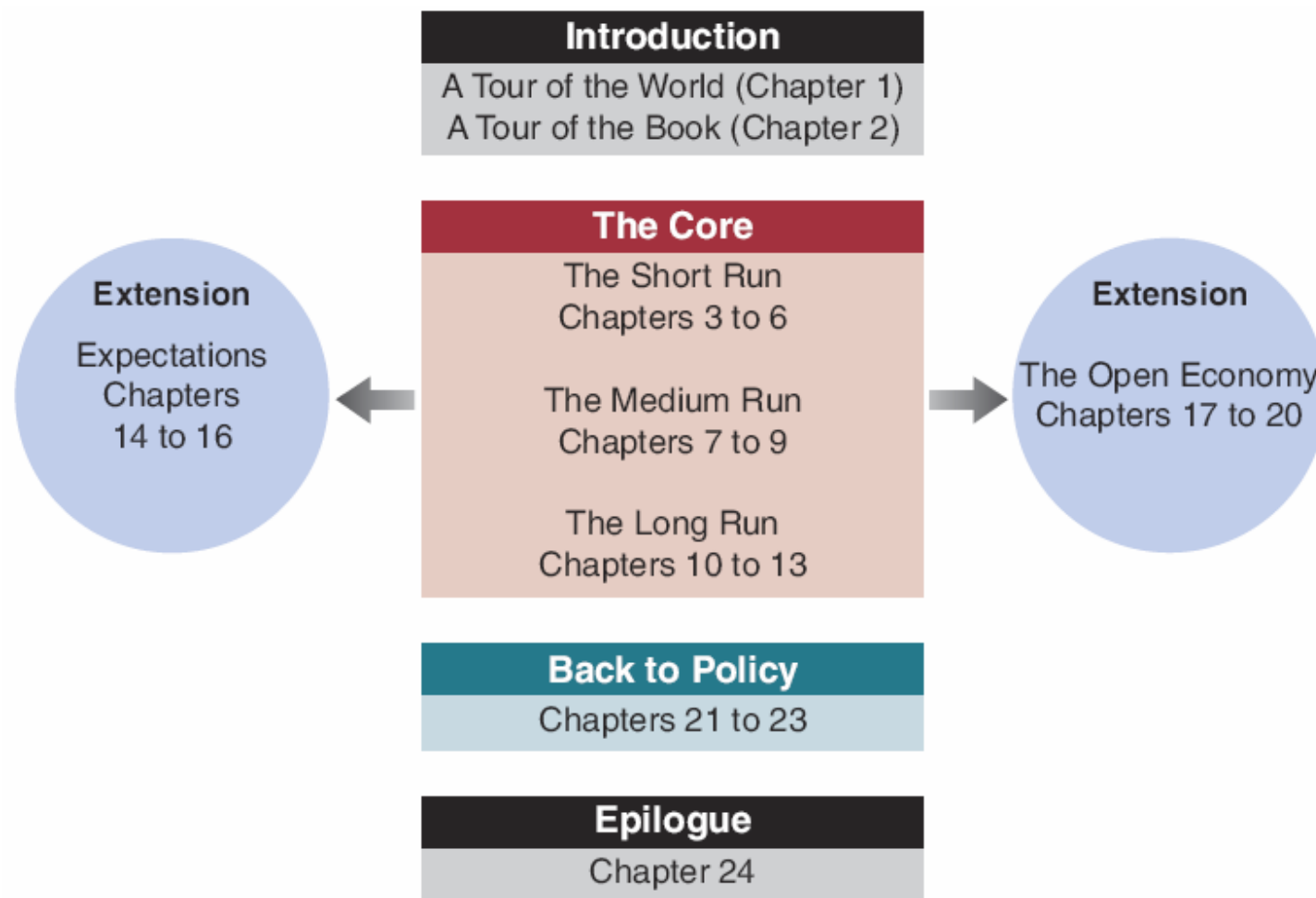
- The Core
 - Chapters 3 to 6: The short run and the role of demand
 - Chapters 7 to 9: The medium run and the supply side
 - Chapters 10 to 13: The long run
- Extensions
 - Chapters 14 to 16: Expectations and implications for fiscal and monetary policy
 - Chapters 17 to 20: Open economy and implications for fiscal and monetary policy

2.6 A Tour of the Book (2 of 3)

- Back to Policy
 - Chapter 21: General issues of policy
 - Chapters 22 & 23: The role of fiscal and monetary policies.
- Epilogue
 - Chapter 24: History of macroeconomics and how macroeconomists have come to believe what they believe today

2.6 A Tour of the Book (3 of 3)

Figure 2.7 The Organization of the Book



APPENDIX: The Construction of Real GDP and Chain-Type Indexes (1 of 3)

- Suppose that an economy produces two final goods, wine and potatoes:

Nominal GDP in Year 0 and in Year 1.

	Quantity	Year 0	
		\$ Price	\$ Value
Potatoes (pounds)	10	1	10
Wine (bottles),	5	2	10
Nominal GDP			20

	Quantity	Year 1	
		\$ Price	\$ Value
Potatoes (pounds)	15	1	15
Wine (bottles),	5	3	15
Nominal GDP			30

- The rate of growth of nominal GDP from year 0 to year 1 is $(\$30 - \$20)/\$20 = 50\%$.

APPENDIX: The Construction of Real GDP and Chain-Type Indexes (2 of 3)

- Suppose year 0 is the **base year**:
 - Real GDP in year 0: $(10 \times \$1) + (5 \times \$2) = \$20$
 - Real GDP in year 1: $(15 \times \$1) + (5 \times \$2) = \$25$
 - The rate of growth of real GDP from year 0 to year 1 is $(\$25 - \$20)/\$20 = 25\%$.
- Suppose year 1 is the base year:
 - Real GDP in year 0: $(10 \times \$1) + (5 \times \$3) = \$25$
 - Real GDP in year 1: $(15 \times \$1) + (5 \times \$3) = \$30$
 - The rate of growth of real GDP from year 0 to year 1 is $(\$30 - \$25)/\$25 = 20\%$.
- Problem: Which base year should one choose?

APPENDIX: The Construction of Real GDP and Chain-Type Indexes (3 of 3)

- In December 1995, the U.S. Bureau of Economic Analysis shifted to a new method with four steps:
 1. Construct the rate of change of real GDP between two years in two different ways:
 - Using the price from year t as the set of common prices
 - Using the price from year $t+1$ as the set of common prices
 2. Construct the rate of change of real GDP as the average of these two rates of change
 3. Construct an index of the level of real GDP by *linking* or *chaining* the constructed rates of change for each year
 4. Multiply this index by nominal GDP to derive *real GDP in chained dollars*